@ create a structure variable student and estimate the size of the structure manually and write a 'c' program to find the size of the structure using size of operator. Now observe both the outputs are same are different. Justify the answer.

Ans:

Structure definition

```
struct student
{
  int rollno;
  char grade;
  float marks;
};
```

Size estimation

int roll = 4 bytes

char grade = 1 byte

float marks = 4 bytes

size = 4 + 1 + 4 = 9 bytes.

C program using sizeof:

```
#include<stdio.h>
struct student
{
 int rollno;
 char grade;
 float marks;
};
int main()
{
 struct student s;
 printf("size=%zu bytes\n", sizeof(s));
 return 0;
}
```

o/p: size = 12 bytes

The o/p are different because compiler adds padding b/w structure members so that data is stored at memory addresses that are efficient for processor to access. Although actual data occupies 9 bytes, compiler inserts 3 extra bytes, making it 12 bytes